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Bridging Side-channel Evaluation Gap: From Interpretable Leakage Models to Concrete Evidence-based Attacks

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Abstract. Side-channel evaluations of cryptographic implementations have been mandated at high security levels. Leakage assessments and attack evaluations are two fundamental aspects. However, there exists a mismatch between leakage assessment outcomes and attack evaluation results. Specifically, popular leakage assessment methods employ well-known statistical tools such as Mutual Information (MI) to quantify the amount of information leaked. Yet, no evidence is provided to further address the more critical problem of whether the theoretically assessed leakages are exploitable in a concrete attack. As is now widely recognized, statistical guidelines offer limited interpretability for resistance against specific attacks. This work bridges this long-standing gap through three major contributions. First, we leverage a family of globally convex/concave and coercive functions to profile a novel noisy leakage model that is grounded in geometric theory. This model enables leakage assessments within an interpretable geometric paradigm. Second, we derive multi-order leakage evidence from the model and demonstrate its exploitability in developing attacks. This is also an evaluation of leakage exploitability. We propose three novel evidence-based distinguishers to validate our claimed exploitability. Third, we formulate worst-case leakage assessments as constrained problems and develop a convergent algorithm to approximate the first-order worst-case leakage bound as an example. The improved distinguisher is derived based on the worst-case evidence. Theoretical analysis has demonstrated that the multi-order leakage evidence is consistently achievable and uniquely characterizable. Furthermore, the convergence rate toward approximating the first-order worst-case bound is ensured to be at least sublinear. Experimental analyses validate the theory, confirming its effectiveness in practical applicability.

Keywords: Side-channel attacks · leakage exploitation · interpretable framework · security bound · convergent approximation

1 Introduction

Side-channel analyses (SCAs) break the traditional black-box assumption of a cryptographic security model by leveraging unintentional leakages from hardware architectures.

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Prominent physical sources include power consumption [KJJ99], electromagnetic radiation [AARR02] and execution time [Koc96], etc. Since such leakages exhibit strong mathematical dependencies on sensitive intermediates, they pose a significant threat to cryptographic implementations. In response, security evaluation frameworks such as Common Criteria (CC) [Com17] and FIPS 140-3 [Inf19] have mandated the evaluation of cryptographic implementations against SCAs at high levels. Among all concerns, how to quantify an implementation's side-channel resistance has become a top one to consider. To this end, security evaluation frameworks have followed two parallel directions: attack evaluations (e.g., Common Criteria [Com17]) and leakage assessments (e.g., FIPS 140-3 [Inf19]).

Attack evaluations. As indicated in the research [SMY09], attack evaluation aims to quantify the practical resistance of cryptographic implementations to side-channel attacks. The criteria are typically evaluated using metrics such as success rate (SR) and guessing entropy (GE). Such evaluations straightforwardly disclose vulnerabilities in the target implementations by determining whether an attack succeeds, and as well quantify the severity of such detected vulnerabilities via checking the conditions (e.g., the number of measurement traces) required for a successful attack. Side-channel distinguishers commonly used in such evaluations include, but are not limited to, correlation power analysis (CPA) [BCO04], partition power analysis (PPA) [LCC+06], template attacks (TA) [CRR02], stochastic attacks (SA) [SLP05] [GLRP06], linear regression analysis (LRA) [MOW17], Bayes Attacks (BA) [HRG14] [GG20], etc. However, it is infeasible to exhaustively evaluate all distinguishers for attack evaluations under limited resources or time constraints.

Leakage assessments. To tackle the issue above, security evaluations have shifted toward determining whether and to what extent observable side-channel leakages reveal information about sensitive intermediate variables [GO24]. The results are used to analyze the mathematical dependencies [DSVC14][SM15] between the observed leakages and the target intermediate variables. Current leakage assessments are based on the fact that an insecure implementation leaks information objectively, regardless of whether a concrete adversary exploits it. As a consequence, the assessment results are adversary-agnostic.

Side-channel security evaluations have evolved over a long period, giving rise to a wide range of attack-based methods and leakage assessment approaches. On the one hand, attack evaluations are specific, they do not rely on explicit leakage models or noise characterizations, since any successful attack against a cryptographic implementation directly reveals specific vulnerabilities in its resistance. However, results from a specific attack evaluation are difficult to generalize to other existing attacks and provide limited guidance on novel attacks. On the other hand, leakage assessments are attack-agnostic and quantify mathematical dependencies to assess theoretical resistance. However, leakage assessments do not provide further evidence on whether the assessed mathematical dependencies are practically exploitable. In the context of growing emphasis on practices, as highlighted in Eurocrypt 24 [GO24], it becomes imperative for leakage assessments to address an open challenge: "*How to leverage the (theoretically) assessed leakages to launch an attack?*"

1.1 Related Works

According to the underlying theories of evaluation models, canonical side-channel security evaluation models can be mainly categorized into four branches as indicated by [CGD22]:

Proof-based model. The proof-based model intends to establish formal security bounds for side-channel resistance of the target cryptographic implementations by employing analytical models such as the probing model [ISW03] and the associated variants [DDF19] [DFS15] [PGMP19]. These models enable formal security proofs under mild assumptions, such as the proportional noise growth hypothesis [CJRR99]. However, as highlighted in [DFS15], the derived security bounds tend to be overly conservative, even in straightforward scenarios involving Hamming weight (HW) leakages with additive Gaussian noise [dCGRP19]. As indicated in [BCGR24], integrating more quantitative analyses into

the assessment model could help refine the proof to achieve tighter security bounds.

Information-theoretic model. The information-theoretic model employs noisiness measures to evaluate side-channel resistance by analyzing profiled leakage distributions. Its objective is to quantify the upper bound of information that an ideal adversary could potentially derive from leakage distributions. By establishing information-theoretic limits on the theoretical success rate, the model can assess worst-case leakage scenarios from an adversary perspective. Commonly used metrics include Shannon conditional entropy, Kullback-Leibler (KL) divergence, Total Variation Distance (TVD) [PW25], Mutual Information (MI) [GBTP08], Euclidean Bias (EB) [PR13], the Complementary Doeblin Coefficient [Dob56], etc. However, the reliability of the noisiness metrics heavily depend on the prior estimation accuracy of the profiled distributions. When only a limited number of traces are available, estimation inaccuracies can distort the final assessment results.

Moment-based model. The moment-based model aims to identify the minimum order of statistical moments in leakages that are significantly correlated with the target sensitive intermediate values. Popular metrics used in the moment-based model include the Signal-to-Noise Ratio (SNR) [SMY09], Analysis of Variance (ANOVA) [TRR+14], Normalized Inter-Class Variance (NICV) [BDGN14], etc. Notably, NICV and ANOVA are connected to SNR, as all of them evaluate the second-order statistical variance of leakages to disclose the corresponding statistical dependency on the keys. Previous research has formally established the relationships between these metrics in [BDGN14] [RBG+16].

Attack evaluation model. The attack evaluation model focuses on quantifying the success rate and guessing entropy of a specific side-channel attack against cryptographic implementations. The evaluation model straightforwardly determines the specific security resistance of a cryptographic implementation by examining the corresponding performance of the success rate and guessing entropy in a concrete attack. However, a specific evaluation result provides limited instructions and references for a generic security evaluation.

Recent work has shifted focus toward explainable assessment models for interactive security evaluation and analytical methods for worst-case security analysis. Specifically, the authors of [GO24] emphasize the importance of interpretability in leakage assessment, arguing that the ultimate goal of an evaluator is to develop a practical attack. To this end, they characterize leakage evidence that explains leaking variables and enables construction of at least one concrete attack vector. However, this established statistical connection between the leakage evidence and attacks is not strong enough: the evidence is primarily used to optimize the efficiency of existing attacks (e.g., DPA and TA), rather than to develop new attacks tailored to the leakage evidence itself. In this work, we therefore characterize a novel type of leakage evidence that not only quantifies leakages but can also be leveraged to develop evidence-based attacks. The authors of [CRBO24] employ the Mutual Information (MI) estimator proposed in [GKOV17] to quantify worst-case leakages under an optimal adversarial model. However, when handling high-dimensional leakage variables (exceeding 30 dimensions), the performance of this estimator may degrade. In contrast, this work proposes an interpretable model combined with an optimization algorithm for worst-case leakage assessment. The proposed model and algorithm effectively accommodate high-dimensional leakage variables while maintaining high evaluation efficiency.

1.2 Our Contributions

In the context of security evaluation, this work leverages geometric theory to address an open challenge highlighted in Eurocrypt 24 [GO24] of bridging the long-standing gap between leakage assessment and attack evaluation. Specifically, this work proposes an interpretable leakage assessment model and introduces a geometry-based mechanism to re-evaluate the exploitability of theoretically assessed leakages for multiple specific attacks. The proposed model enables evaluators to connect interpretable leakage assessment with specific attack evaluation, thus providing deeper and more informative insight into abstract

leakage behavior. Considering the ultimate goal of security evaluation, such improvement helps distinguish exploitable and non-exploitable theoretical dependencies from practical security concerns. From a developmental standpoint, this work represents a meaningful step toward achieving a more accurate and finer-grained side-channel security evaluation. For clarity, the major contributions are summarized as follows.

- We leverage a family of convex/concave and coercive functions to profile an interpretable noisy leakage model. The proposed model is grounded in geometric theory, with its core innovation being the capacity to characterize multi-order leakage evidence that admits intuitive geometric interpretation. This advancement enables side-channel leakage assessment within an interpretable paradigm. Theoretically, the multi-order leakage evidence has been proven to be consistently achievable (refer to Expression (3)) and uniquely characterizable (refer to Lemma 1).
- We demonstrate the exploitability of the multi-order leakage evidence by developing concrete attacks. This is essentially an evaluation of leakage exploitability. Specifically, we leverage discrete sequence-limit analysis to anchor the multi-order leakage evidence at the asymptotic minimum (Section 3.3). Furthermore, we propose three novel evidence-based distinguishers to practically validate this exploitability.
- From an optimization-theoretic perspective, we show that identifying the worst-case leakage evidence, i.e., approximating the tightest leakage bound, is challenging. We formulate worst-case leakage assessment as linearly constrained problems and develop an optimization algorithm to identify the first-order worst-case leakage evidence as a representative case. The convergence rate is ensured to be at least sublinear. By leveraging the established connection between leakage assessment and attack evaluation, we derive an improved distinguisher based on the worst-case evidence.

2 Preliminaries

2.1 Side-channel Leakages

Side-channel leakages refer to physical measurements that contain information of sensitive intermediates, thereby enabling potential recovery attacks. In cryptographic implementations, S-Boxes are particularly vulnerable to leakages, since they rely on look-up operations. Let n denote the input bit length of an S-box (e.g., $n = 8$ in the AES and $n = 4$ in the PRESENT). For each block $l_i \in L = \{1, \cdots, |L|\}$, let $K_*^{(l_i)} = \{0,1\}^n \in \mathbb{F}_2^n$ denote the l_i-th key block and $K_o^{(l_i)}$ any possible hypothesis thereof. Let $t_q^{(l_i)} \in \mathbb{F}_2^n$ denote the l_i-th plaintext block encrypted in the q-th implementation where $q \in Q = \{1, \cdots, |Q|\}$. For each pair $(l_i, q) \in L \times Q$, the corresponding side-channel leakages $X_q^{(l_i)}$ are modeled as:

$$X_q^{(l_i)} = \mathcal{G} \circ \mathcal{S}(t_q^{(l_i)} \oplus K_*^{(l_i)}) + \varepsilon, \quad q \in Q, \quad l_i \in L, \tag{1}$$

where $\varepsilon \sim N\left(0, \sigma^2\right)$ is independent Gaussian noise. Here, the S-box $\mathcal{S} : \mathbb{F}_2^n \to \mathbb{F}_2^n$ is a bijective mapping, and the function $\mathcal{G} : \mathbb{F}_2^n \to \mathbb{R}$ models the deterministic physical leakage according to the S-box output, which depends on the underlying hardware architecture. In the sequel, we use $Z_q^{(l_i)} = \mathcal{G} \circ \mathcal{S}(t_q^{(l_i)} \oplus K_*^{(l_i)})$ to denote noise-free leakages, i.e., the full-leakage variable. By profiling $Z_q^{(l_i)}$, an adversary can launch effective attacks against cryptographic implementations, as these quantities contain rich information about the key (see Lemmas 2–5 in [LOW+24]). Let $(\cdot)^\top$ denote the transpose operator, we define the multivariate variables as follows. Let $\vec{K}_* = [K_*^{(l_1)}, \cdots, K_*^{(l_{|L|})}]^\top$ denote the key vector, let $\vec{K}_o = [K_o^{(l_1)}, \cdots, K_o^{(l_{|L|})}]^\top$ denote any key hypothesis, and let $\vec{X}^{(l_i)} = [X_1^{(l_i)}, \cdots, X_{|Q|}^{(l_i)}]^\top$ represent the $|Q| \times 1$ leakage column vector of the l_i-th block over $|Q|$ encryptions.

2.2 Metrics in Attack Evaluations

Success rate (SR) constitutes a popular criterion in evaluating side-channel attacks, which represents the probability that an adversary successfully recovers the key [SMY09]. There exist multiple orders of SR pertaining to post-processing key enumerations after launching a concrete attack. Let $Rank$ be the ranking function, and Pr be the probability function. In the sequel, we consider the most straightforward and widely used first-order SR, i.e.,

$$SR = Pr\{Rank(\vec{K}_*) = 1\}.$$

This expression models the probability of the key being ranked as the best candidate. Evaluating the theoretical SR is challenging, as it requires stringent assumptions and conditions. Although several studies have attempted to establish bounds on SR, these bounds are often loose in practical validations. The most common approximation to estimate SR is to adopt an empirical sampling, whereby multiple practical runs are performed to approximate $\hat{SR} \to SR$. As a complementary metric, guessing entropy (GE) [SMY09] provides an informative criterion by evaluating the expected rank of the key $\vec{K}_*$ among all possible hypothesis $\vec{K}_o$, i.e.,

$$GE = E(Rank(\vec{K}_*)).$$

Likewise, an empirical estimate of GE is used to realize $\hat{GE} \to GE$. GE complements SR through revealing how incorrect hypotheses behave before any successful attack occurs.

3 From Interpretable Models to Exploitable Evidence

Our assessment model aims to assess any key-related leakages and we thus use an abstract function $F(\vec{K}_*)$ to represent any key-related variable. Specifically, when F is the identity function, $F(\vec{K}_*) = \vec{K}_*$. In the sequel, $F(\vec{K}_*)$ and $F(\vec{K}_o)$ are abbreviated as F_* and F_o.

3.1 Interpretable Leakage Assessment Model

Firstly, we define an *interpretable* assessment model: an assessment model is *interpretable*, if it measures similarity between leakages $\mathcal{X}$ and $\mathcal{Z}$ (where $\mathcal{Z}$ is approximated by the profiled $\hat{\mathcal{Z}}$ in practice) using geometric metrics. This *interpretability* stems from a straightforward geometric mechanism. Specifically, according to geometric theory, points at special positions on a given geometry possess special properties that distinguish them from ordinary points. These special points include extremal points, stationary points and saddle points, etc. They often mark pivotal transitions in geometric behavior, such as monotonicity, concavity and convexity, etc. Therefore, the mechanism lies in: by projecting leakages Z onto these special points, the geometric embedding of Z is endowed with these unique geometric properties. Here, the geometric embedding refers to a structured characterization of leakages in a given geometric space. In this sense, any similar properties detected in leakages X are straightforward evidence of its geometric similarity to leakages Z (or profiled $\hat{\mathcal{Z}}$).

Since extremal points exhibit the most elementary geometric properties with reduced analytical complexity, we opt for extremal points as projection targets. The corresponding geometry-wise properties thus include the first-order condition where the function value is a maximum or minimum, the second-order condition where the gradient vector vanishes, the third-order condition where the Hessian matrix is positive semi-definite or negative semi-definite, etc. To ensure the existence of extremal points and the uniqueness of extrema, it is obligatory to establish the assessment model via a family of global convexity/concavity and coercive functions, as indicated by Weierstrass theorem (Section 2, [BNO03]).

Formally, we demonstrate the interpretable assessment model in Definition 1. In Lemma 1, we prove that the proposed model admits a unique characterization, i.e., the characteristics of the model cannot be misidentified due to its convexity/concavity property.

Definition 1. Let $\mathcal{X}_L = \{\vec{X}^{(l_1)}, \vec{X}^{(l_2)}, ..., \vec{X}^{(l_{|L|})}\}$ represent a random vector sequence of leakages on a discrete set X and $\mathcal{V}$ denote the leakage space. Let $dom(\mathcal{D})$ denote the domain of function $\mathcal{D}$, and Θ denote an embedding factor (either a vector or a matrix). For a globally concave or convex, coercive and stochastic function $\mathcal{D} : dom(\mathcal{D}) \to \mathbb{R}$, if

$$\begin{cases} \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}_{max}| \leq \zeta, & \mathcal{D} \text{ is concave,} \\ \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}_{min}| \leq \zeta, & \mathcal{D} \text{ is convex,} \end{cases} \tag{2}$$

where the infimum is taken over all sets $X \subseteq \mathcal{V}$ and $\mathcal{I}$ is a nested function, we say that the bound ζ constitutes the first-order leakage evidence of $\mathcal{X}_L$. We also say that $\mathcal{X}_L$ is characterized $(\Theta, \zeta)_{\mathcal{D}}$-GD-noisy leakage from F_*, where 'GD' denotes geometric distance. Here $\mathcal{D}_{max/min} = \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot)$ represents a constant extremal contour independent of the embedding parameter Θ, which we refer to as the full-leakage contour.

It is noteworthy that $\mathcal{D}$ acts as a leakage evaluator in the interpretable model as given in Expression (2). In the context of leakage resistance, if a cryptographic implementation is said to be resistant, there exist some sufficiently large bounds $res \to +\infty$, such that:

$$\inf_{(\mathcal{D},\Theta)} \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}_{max/min}| \geq res. \tag{3}$$

Indeed, satisfying the above inequality (3) in practice is challenging. Therefore, there always exists a finite positive constant ζ that serves as an upper bound for Expression (2). This follows that if a cryptographic implementation suffers from GD-noisy leakage, such leakage can always be captured or identified by an appropriate $\mathcal{D}$. This indicates that the leakage evidence ζ of GD-noisy leakage is consistently achievable.

Lemma 1. *For the first-order evidence ζ and the $(\Theta, \zeta)_{\mathcal{D}}$-GD-noisy leakage model, there exists no other constant extremal contour $\mathcal{D}^o \neq \mathcal{D}_{max/min}$ within some neighborhoods $\mathcal{V}_{nei} \subseteq \mathcal{V}$, and no other different upper bound $\zeta^o \neq \zeta$, such that:*

$$\begin{cases} \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}^o| \leq \zeta^o, & \mathcal{X}_L \in \mathcal{V}_{nei}, \\ \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}^o| \leq \zeta^o, & \mathcal{X}_L \in \mathcal{V}_{nei}. \end{cases}$$

Proof. Consider a globally convex $\mathcal{D}$ and let $\mathcal{D}^o$ denote an extremal contour of $\mathcal{D}$. For any neighborhood $\mathcal{V}_{nei} \subseteq \mathcal{V}$ and for any $\mathcal{Z}_L^{(nei)} \in \mathcal{V}_{nei}$, it holds that $\mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L^{(nei)}); \cdot) \geq \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) = \mathcal{D}^o$. Suppose toward a contradiction that there exists $\widetilde{\mathcal{Z}_L} \in \mathcal{V}_{nei}$ such that $\mathcal{D}(\mathcal{I}(F_*; \widetilde{\mathcal{Z}_L}); \cdot) < \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) = \mathcal{D}^o$. In this case, we consider a convex combination $\mathcal{Z}_L(\rho) = \rho\mathcal{Z}_L + (1-\rho)\widetilde{\mathcal{Z}_L}$ for $\rho \in [0, 1]$. By the convexity of $\mathcal{D}$, it holds that

$$\begin{aligned} \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L(\rho)); \cdot) &\leq \rho \cdot \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) + (1-\rho) \cdot \mathcal{D}(\mathcal{I}(F_*; \widetilde{\mathcal{Z}_L}); \cdot) \\ &< \rho \cdot \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) + (1-\rho) \cdot \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) \\ &= \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot), \quad \text{for } \rho \in [0, 1]. \end{aligned}$$

For any limit $\rho \to 1 \in [0, 1]$ such that $\mathcal{Z}_L(\rho) \in \mathcal{V}_{nei}$, we have $\mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L(\rho)); \cdot) \geq \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) = \mathcal{D}^o$. This instantly leads to a contradiction. Therefore, we have $\mathcal{D}^o \leq \mathcal{D}(\mathcal{I}(F_*; \widetilde{\mathcal{Z}_L}); \cdot)$ for all $\mathcal{Z}_L \in \mathcal{V}$, indicating that $\mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L^{(nei)}); \cdot) \geq \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot) = \mathcal{D}^o$ for all $\mathcal{Z}_L^{(nei)} \in \mathcal{V}$. This also implies that $\mathcal{D}^o \equiv \mathcal{D}_{min}$, which consequently leads to $\zeta^o = \zeta$ in Expression (2). For a globally concave $\mathcal{D}$, the proof follows a similar line of reasoning. □

Lemma 1 guarantees that for any given factor Θ, the evidence ζ is unique and cannot be misidentified, therefore preventing a homogeneous issue presented in [IUH22][1].

[1]The research in [IUH22] reports a non-unique dependency between the Cross-entropy (CE) / Perceived Information (PI) metrics and the associated probability distributions, as stated in Proposition 3.

In the context of noisy leakage models, we compare $(\Theta, \zeta)_{\mathcal{D}}$-GD-noisy leakage with ($t$, δ)-SD-noisy leakage (including its generalized version (t, δ)-GSD-noisy leakage) discussed in Crypto 24 [ORR+24], to provide an interactive understanding. The latter model adopts statistical distance (SD) for noisy leakage model characterization, i.e.,

$$\sup_{S}[P_{XZ}(S) - 2^t \cdot P_X(S) \otimes P_Z(S)] \leq \delta, \tag{4}$$

where $\mathrm{SD}_t(P;Q) = \sup_S[P(S) - 2^t \cdot Q(S)]$ denotes the t-hockey-stick divergence [SV16] between the distributions P and Q. It is not hard to note that the factor 2^t in Expression (4) aligns with the role of the embedding factor Θ in Expression (2), and δ, ζ respectively serve as upper bounds for each measurement. The proposed noisy leakage model given in Expression (2) therefore shares similarities with the core model in Expression (4) of Crypto 24 [ORR+24] in terms of both analytical bounds and approximation theory. If a statistical distribution of $\mathcal{D}$ is established, Expression (2) can be seamlessly reformulated in the form of statistical distance. The critical differences lie in the noisiness metrics: (I) (t, δ)-SD-noisy leakage (including its variant (t, δ)-GSD-noisy leakage) is irreversible or asymmetric due to the fact $\mathrm{SD}_t(P;Q) \neq \mathrm{SD}_t(Q;P)$; (II) $(\Theta, \zeta)_{\mathcal{D}}$-GD-noisy leakage possesses higher-order features according to Taylor's theorem. In this sense, the advantage of characterizing GD-noisy leakage is straightforward: the leakage evidence has the ability to extract more complex and higher-order mathematical dependencies from F_*, which helps reveal more nuanced sensitive information compared to traditional single-order counterparts.

It is worth mentioning that the Gaussian Mixture Model (GMM) [LRP07] is a widely used technique in leakage assessment, as it provides a flexible tool for approximating leakage distributions. However, since a GMM is primarily used for distribution fitting, it may not explain why theoretically assessed leakages are exploitable in concrete attacks. In this regard, a GMM may not provide the level of interpretability pursued in this work.

3.2 Higher-Order Leakage Evidence

In light of Taylor's theorem, the proximity between a neighboring noisy leakage sequence $\mathcal{X}_L$ and a centered constant leakage sequence $\mathcal{Z}_L$ depends on information across all orders. The proposed interpretable assessment model thus benefits from using multi-order geometric metrics to quantify richer geometry-wise similarity. The high-order geometric metrics used in this work are straightforwardly selected as order-based differences[2], which equally reflect the homogeneous properties on $\mathcal{D}$. These differences are further reduced to scalar bounds via the Euclidean vector norm $\|\cdot\|$ and the Frobenius matrix norm $\|\cdot\|_{\mathbf{F}}$. Formally, let ζ_2 and ζ_3 denote the second-order and third-order leakage evidence, if they upper-bound:

$$\begin{cases} \inf_X \|\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)\| \leq \zeta_2, \\ \inf_X \|\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta) - \nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)\|_{\mathbf{F}} \leq \zeta_3, \end{cases} \tag{5}$$

where $\nabla\mathcal{D}$ is the gradient vector and $\nabla^2\mathcal{D}$ is the Hessian matrix. We say that $\mathcal{X}_L$ is characterized higher-order $(\Theta, \zeta, \zeta_2, \zeta_3)_{\mathcal{D}}$-GD-noisy leakage from F_*. Theoretically, the numerical relationships between the evidence ζ, ζ_2 and ζ_3 are given in Theorem 1.

Theorem 1. *For the $(\Theta, \zeta, \zeta_2, \zeta_3)_{\mathcal{D}}$-GD-noisy leakage model, if the nested function $\mathcal{I} \in dom(Conv)$, the numerical relationships between the evidence ζ, ζ_2 and ζ_3 follow that:*

$$\zeta = \mathcal{O}(\zeta_2\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|) \simeq \mathcal{O}(\zeta_2), \tag{6}$$

$$\zeta_2 = \mathcal{O}((\zeta_3 + \lambda_{max}[\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)])\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|) \simeq \mathcal{O}(\zeta_3). \tag{7}$$

Here, $dom(Conv)$ denotes the domain of convergence, $\mathcal{O}(\cdot)$ is Landau's notation, and $\lambda_{max}[\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)]$ represents the corresponding largest eigenvalue(s).

[2]Alternatives also exist, e.g., the condition number $\lambda_{\max}/\lambda_{\min}$ of the second-order Hessian matrix.

Proof. The proof is deferred to Appendix A for readability. □

It should be noted that the characterized leakage evidence ζ, ζ_2, and ζ_3 corresponding to different Taylor orders do not contain overlapped information, i.e., they are mutually orthogonal. Taking a one-dimensional $\mathcal{D}$ as an example, the value of $\mathcal{D}$ at a given point is independent of both the slope of its tangent line and its local concavity at that point.

3.3 Exploiting Leakage Evidence for Attack Distinguishers

With the interpretable model introduced in Section 3.1, here we further define *exploitable* leakage evidence: if there exists at least one attack distinguisher that can exploit the assessed leakage evidence, the evidence is *exploitable*. Notice that the leakage evidence here is the assessment criterion derived from the *interpretable* model given in Expression (2).

Thanks to the model's interpretability, it is easy to develop distinguishers by exploiting the leakage evidence ζ. The mechanism is based on the fact that as the degree of leakage increases, the evidence ζ is asymptotically minimized at F_*, i.e., $\zeta(F_*) = \min_{F_o}\{\zeta(F_o)\}$. The theoretical form of an distinguisher is readily derived as $F = \operatorname{argmin}_{F_o} \zeta(F_o)$. Notice in Expression (2), the nested function $\mathcal{I}(F_*; \mathcal{X}_L)$ is a dependency measure between F_* and leakages $\mathcal{X}_L$. It follows that, if this dependency is especially quantified by information tools[3], the function $\mathcal{D}$ becomes an attack distinguisher with a soundness guarantee [SMY09] [GHR15]. A proof is provided as follows. For $|Q| \to +\infty$ (i.e., given enough number of traces), it holds that $\mathcal{I}(F_*; \mathcal{X}_L) \to \mathcal{I}(F_*; \mathcal{Z}_L)$, which implies $\zeta(F_*) \to 0$, it arrives at:

$$\lim_{|Q|\to+\infty} SR(\mathcal{X}_L) = \lim_{|Q|\to+\infty} Pr\{\zeta(F_*) \leq \zeta(F_o)|\mathcal{X}_L\} = 1. \tag{8}$$

In fact, any unbiased statistical estimator is a feasible alternative for $\mathcal{I}$. The workflow of developing distinguishers exploiting higher-order evidence ζ_2 and ζ_3 is analogous.

An important advantage of developing attacks based on *exploitable* evidence lies in that, since the attack leverages all the assessed evidence, the evidence, in turn, quantifies the degree of leakages used by the attack. A concise proof is given as follows. For a random evidence variable $\zeta(F_o)$ with the probability density function denoted by $f_\zeta(x)$, we have:

$$\begin{aligned} SR(\mathcal{X}_L) &= Pr\{\zeta(F_*) \leq \zeta(F_o)\} = \int_{\zeta(F_*)}^{+\infty} f_\zeta(x)dx, \\ GE(\mathcal{X}_L) &= \sum_i i \cdot Pr\{Rank(\zeta(F_*)) = i\} = \sum_i i \cdot Pr\{\zeta(F_*) \leq \Omega_i, \zeta(F_*) > \overline{\Omega}_i\}, \end{aligned} \tag{9}$$

where Ω_i denotes the discrete set of the first i smallest $\zeta(F_o)$ excluding $\zeta(F_*)$, and $\overline{\Omega}_i$ represents its complementary set. The condition $\zeta(F_*) \leq \Omega_i$ indicates that the variable $\zeta(F_*)$ is less than any element in the set Ω_i. The other condition $\zeta(F_*) > \overline{\Omega}_i$ follows the same convention. Expression (9) implies that stronger evidence, i.e., a smaller value of $\zeta(F_*)$, leads to higher SR and lower GE. This observation is informative: a generally higher degree of leakages does not necessarily correspond to better SR or GE, as attack efficiency also depends on distinguisher's structure. Instead, SR increases and GE decreases only when the degree of exploitable leakage in a specific attack increases.

For clarity, we elaborate on the mechanism for transforming leakage evidence into distinguishers as follows. Due to $\lim_{q\to\infty} \mathcal{I}(F_*; \mathcal{X}_L) = \mathcal{I}(F_*; \mathcal{Z}_L)$, it is sufficient to have:

$$\lim_{q\to\infty} \inf_X |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}(\mathcal{I}(F_*; \mathcal{Z}_L); \cdot)| = \lim_{q\to\infty} \inf \zeta_*(q) = 0, \quad q \in Q.$$

The inferior limit $\liminf_{q\to\infty} \zeta_*(q)$ can be equally expressed as $\lim_{q\to\infty} \inf_{k\geq q} \zeta_*(k)$. In the case $F = F_o$, it holds that $\lim_{q\to\infty} \inf_{k\geq q} \zeta_o(k) > 0$. Let $\{\zeta_*(1), \cdots, \zeta_*(k)\}_{k\in Q}$ and

[3]Most basically, consider variables V_A and V_B with the same amount of information: $\mathcal{I}(V_A) = \mathcal{I}(V_B)$.

$\{\zeta_o(1), \cdots, \zeta_o(k)\}_{k\in Q}$ denote two discrete evidence sequences indexed by $k \in Q$. The fact $\lim_{q\to\infty} \inf_{k\geq q} \zeta_o(k) > \lim_{q\to\infty} \inf_{k\geq q} \zeta_*(k)$ implies, by the preservation property of sequence limits (Section 3, [Rud76]), that there exists a finite threshold $|Q|_{thr} > 0$ such that for any $k > |Q|_{thr}$, the inequality $\zeta_*(k) < \zeta_o(k)$ holds strictly. Therefore, the leakage evidence ζ_* is uniquely distinguished as the asymptotic minimum among hypotheses. Three evidence-based distinguishers are given in Section 4 to validate this exploitability.

4 Concrete Side-channel Distinguishers

We propose three distinguishers based on the first-order, second-order, and third-order leakage evidence. The sensitive function F_* is instantiated either as the key $F(\vec{K}_*) = \vec{K}_*$ or as a collision chain, depending on the evaluation scenario. Denote the i-th entry of F_* as $F_*\{i\}$, the collision chain satisfies $F_*\{i\} \oplus F_*\{j\} = K_*^{(l_i)} \oplus K_*^{(l_j)}$, for any blocks $l_i, l_j \in L$.

4.1 Three Novel Attack Distinguishers

We present the first distinguisher in Proposition 1. For a concave function, the formulation of a distinguisher $F_* = \text{argmin}_{F_o} \zeta(F_o)$ is equivalent to $F_* = \text{argmax}_F \mathcal{D}(\mathcal{I}(F; \mathcal{X}_L); \Theta)$. When F_* is the key, we denote U_{pro} as the profiling set and X_{pro} as the profiled leakages. When F_* is a collision chain, we denote $U_{i,j}$ as the set of hypothesis collisions between the l_i-th and l_j-th blocks, that is, for any $u_{i,j} \in U_{i,j}$, the hypothesis collision leads to the equation $t_q^{(l_i)} \oplus K_o^{(l_i)} = t_q^{(l_j)} \oplus K_o^{(l_j)} = u_{i,j} \in U_{i,j}$. For clarity, we denote the entry $F\{i\}$ $(1 \leq i \leq |L|)$ as the origin of a hypothesis chain F, and $F\{i\}$ becomes the only free variable in F. Let $\|\cdot\|_1$ denote the ℓ_1-norm and $|\cdot|$ denote the cardinality of a finite set.

Proposition 1. *For any $|U_{i,j}| \times |U_{i,j}|$ diagonal matrix $diag(\omega_{i,j})$ with positive diagonal entries satisfying $\sum \omega_{i,j} = 1$, for any $|U_{i,j}| \times 1$ constant drift vector $\vec{C} = [c]^\top$ with $c > 0$, and let $\Theta = [diag(\omega_{i,j}), \vec{C}]$. A concrete vector-variate distinguisher $\mathcal{D}_{o_1}(\hat{\mathcal{I}}(F; \mathcal{X}_L); \Theta)$, defined on $\mathcal{R}^{|U_{i,1}|} \times \cdots \times \mathcal{R}^{|U_{i,|L|}|} \to \mathcal{R}$, can take the expression:*

$$
\begin{aligned}
F_{output}^{o_1} &= \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\text{argmax}}\, \mathcal{D}_{o_1}(\hat{\mathcal{I}}(F; \mathcal{X}_L); \Theta) \\
&= \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\text{argmax}} \sum_{l_j\in L/\{l_i\}} \prod_{u_{i,j}\in U_{i,j}} (\bar{X}_{st}^{(l_i)}/\bar{X}_{st}^{(l_j)} + c)^{\omega_{i,j}} \\
&\qquad - \sum_{l_j\in L/\{l_i\}} \left\| diag(\omega_{i,j}) \cdot [\bar{X}_{st}^{(l_i)}/\bar{X}_{st}^{(l_j)} + c]^\top \right\|_1,
\end{aligned}
\tag{10}
$$

where $\widehat{\mathcal{I}}(F; \mathcal{X}_L) = [\bar{X}_{st}^{(l_i)}/\bar{X}_{st}^{(l_j)}]^\top$ is defined on $(\mathbb{F}_2^n)^{|L|} \to \mathcal{R}^{|U_{i,1}|} \times \cdots \times \mathcal{R}^{|U_{i,|L|}|}$, and $\bar{X}_{st}^{(l_i)}$ is given by the expression: $\bar{X}_{st}^{(l_i)} = \sum_{q=1}^{|Q|} X_q^{(l_i)} / \sum_{q=1}^{|Q|} 1$, s. t. $t_q^{(l_i)} \oplus F\{i\} = u_{i,j} \in U_{i,j}$.

Proposition 1 introduces an evidence-based distinguisher for the security evaluation scenario where F_* is the collision chain. When F_* is targeted as the key $\vec{K}_*$, the term $\bar{X}_{st}^{(l_i)}/\bar{X}_{st}^{(l_j)}$ in Expression (10) is replaced by $\bar{X}_{st}^{(l_i)}/X_{pro}$ with $\bar{X}_{st}^{(l_i)} = \sum_{q=1}^{|Q|} X_q^{(l_i)} / \sum_{q=1}^{|Q|} 1$, s. t. $Im(\vec{K}_*) \in U_{pro}$, where the term $Im(\vec{K}_*)$ denotes a key-related sensitive intermediate.

It is worth noting that the embedding factor Θ s a column-augmented matrix composed of a constant drift vector and a diagonal sparse matrix. Following the same formulation, the other two evidence-based distinguishers, $\mathcal{D}_{o_2} = \text{argmin}_{F\in(\mathbb{F}_2^n)^{|L|}} \zeta_2(F)$ and $\mathcal{D}_{o_3} = \text{argmin}_{F\in(\mathbb{F}_2^n)^{|L|}} \zeta_3(F)$, are formally defined in Expressions (11) and (12), respectively. To reduce the computational complexity associated with matrix algebra in $\mathcal{D}_{o_3}$, we intentionally

reformulate $\mathcal{D}_{o_3}$ into a finite-sum form. The high-order distinguishers are as follows:

$$F^{o2}_{output} = \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\operatorname{argmin}}\ \zeta_2(F) = \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\operatorname{argmin}}\ \|\nabla\mathcal{D}_{o_1}(\hat{\mathcal{I}}(F;\mathcal{X}_L);\Theta)\|, \tag{11}$$

$$F^{o3}_{output} = \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\operatorname{argmin}}\ \zeta_3(F) = \underset{F\in(\mathbb{F}_2^n)^{|L|}}{\operatorname{argmin}} \sum_{l_j\in L/\{l_i\}} \|\nabla^2\mathcal{D}^{(i,j)}_{o_1} - \mathcal{M}^{(i,j)}\|_{\mathbf{F}}, \tag{12}$$

where $\mathcal{D}^{(i,j)}_{o_1}$ denotes the j-th addictive item in $\mathcal{D}_{o_1}$, i.e.,

$$\mathcal{D}^{(i,j)}_{o_1} = \prod_{u_{i,j}\in U_{i,j}} (\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st} + c)^{\omega_{i,j}} - \left\| diag(\omega_{i,j}) \cdot [\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st} + c]^{\top} \right\|_1. \tag{13}$$

Let us denote the i-th entry of $[\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st} + c]^{\top}$ by $\psi\{i\}$, and the k-th diagonal entry of $diag(\omega_{i,j})$ by $\epsilon\{k\}$. The term $\mathcal{M}^{(i,j)}$ is a square constant matrix with its (m,m)-th main diagonal entry being $\prod_{u_{i,j}\in U_{i,j}}(\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st}+c)^{\omega_{i,j}}\cdot(\epsilon\{m\}^2-\epsilon\{m\})/\psi\{m\}^2$, and its (m,n)-th non-diagonal entry being $\prod_{u_{i,j}\in U_{i,j}}(\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st}+c)^{\omega_{i,j}}\cdot\epsilon\{m\}\epsilon\{n\}/(\psi\{m\}\psi\{n\})$. In the security evaluation scenario where F_* is targeted as the key $\vec{K}_*$, the term $\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st}$ in Expressions (12) and (13) is replaced by the profiled term $\bar{X}^{(l_i)}_{st}/X_{pro}$.

4.2 Formal Proof of the Conditions on $\mathcal{D}_{o_1}$

To guarantee the geometric interpretability of the proposed assessment model and the exploitability of the multi-order leakage evidence, we formally prove the coercivity and global concavity of the function $\mathcal{D}_{o_1}$, provided in Expression (10), in Theorem 2.

Theorem 2. *The distinguisher $\mathcal{D}_{o_1}$ is coercive and globally concave with respect to $\widehat{\mathcal{I}}$.*

Proof. The function D_{o_1} is constant and reduces to 0 on the diagonal. Let $d^{\perp}$ denote the subspace orthogonal to the diagonal. Along all directions orthogonal to the diagonal, D_{o_1} is coercive that satisfies $D_{o_1}(\widehat{\mathcal{I}}) \to \infty$ as $\|\widehat{\mathcal{I}}\|_2 \to \infty$, for $\widehat{\mathcal{I}} \in d^{\perp}$. Hence, there exists $R>0$ such that $D_{o_1}(\widehat{\mathcal{I}}) < D_{o_1}(0)$ for any $\widehat{\mathcal{I}} \in d^{\perp}$ with $\|\widehat{\mathcal{I}}\|_2 > R$. Therefore, every maximizer of D_{o_1} on $d^{\perp}$ lies in the compact set $\overline{B}_R \cap d^{\perp}$. Since D_{o_1} is also upper semicontinuous, it attains its maximum on $\overline{B}_R \cap d^{\perp}$. Therefore, the set of extremal points of D_{o_1} is nonempty. Based on the concavity-preserving property, the proof of concavity for $\mathcal{D}_{o_1}$ can be reduced to proving that $\mathcal{D}^{(i,j)}_{o_1}$ in Expression (13) is concave in $\widehat{\mathcal{I}} \in \mathcal{R}^{|U_{i,j}|}$. Let $\psi\{i\}$ denote the i-th entry of $[\bar{X}^{(l_i)}_{st}/\bar{X}^{(l_j)}_{st} + c]^{\top}$, and $\epsilon\{k\}$ denote the k-th diagonal entry of $diag(\omega_{i,j})$. By taking the derivative of Expression (13), the first-order $\nabla\mathcal{D}^{(i,j)}_{o_1}$ takes the expression of:

$$\nabla\mathcal{D}^{(i,j)}_{o_1} = [\epsilon\{k\}((\psi_k)^{-1}\prod_k(\psi_k)^{\epsilon\{k\}} - 1)]^{\top}, \quad k = 1,\cdots,|U_{i,j}|.$$

Consequently, the second-order Hessian matrix $\nabla^2\mathcal{D}^{(i,j)}_{o_1}$ is given by the form of:

$$\nabla^2\mathcal{D}^{(i,j)}_{o_1} = -\prod_k(\psi_k)^{\epsilon\{k\}}(diag(\epsilon\{k\}/\psi_k^2) - \vec{P}^{\top}\vec{P}), \quad k = 1,\cdots,|U_{i,j}|.$$

where $\vec{P} = [\epsilon\{k\}/\psi_k]^{\top}$. Consider any arbitrary vector $\vec{V} \in \mathcal{R}^{|U_{i,j}|}$ and denote the i-th entry of $\vec{V}$ by V_i. By using the condition $\sum\omega_{i,j} = \sum_k\epsilon\{k\} = 1$, it is adequate to have:

$$\begin{aligned}\vec{V}^{\top}\nabla^2\mathcal{D}^{(i,j)}_{o_1}\vec{V} &= -\prod_k(\psi_k)^{\epsilon\{k\}}\vec{V}^{\top}(diag(\epsilon\{k\}/\psi_k^2) - \vec{P}^{\top}\vec{P})\vec{V} \\ &= -\prod_k(\psi_k)^{\epsilon\{k\}}(\sum_m\frac{V_m^2\epsilon\{m\}}{\psi_m^2}\cdot\sum_k\epsilon\{k\} - (\sum_n\frac{V_n\epsilon\{n\}}{\psi_n})^2) \le 0,\end{aligned}$$

where the last inference uses Cauchy-Schwarz inequality. Therefore, the Hessian matrix $\nabla^2\mathcal{D}^{(i,j)}_{o_1}$ is negative semi-definite, which implies that $\nabla^2\mathcal{D}_{o_1}$ is negative semi-definite. □

5 Optimization

In this section, we tackle the last issue: "*Does my model assess worst-case leakage evidence?*" Or *"Do I achieve the best achievable*[4] *distinguisher based on current leakage evidence?*" In Section 3, we established an exploitability pathway from leakage evidence to distinguishers. Now, we advance the work to explore a more complex challenge: how to develop an *optimization* pathway from worst-case leakage evidence to improved distinguishers.

According to the conclusions in Section 3, multi-order leakage evidence can be derived via Expressions (2) and (5). However, such evidence does not necessarily constitute the tightest upper bound for Expressions (2) and (5) given the leakages $\mathcal{X}_L$. Assessing the tightest leakage bound is equivalent to identifying an infimum ζ_{wor} that strictly satisfies $\zeta_{wor} \leq \zeta$. We clarify that deriving leakage evidence is not difficult, whereas approximating worst-case leakage bounds is challenging and requires an optimization technique.

5.1 Why to Assess Worst-case Leakages

The notion of an *ideal* adversary was discussed in Cryto 24 [CRBO24]. An adversary is viewed as *ideal* if they have complete knowledge of leakages. To understand the genuine security of a cryptographic device or implementation, the authors of [CRBO24] argue that an assessment model must (approximately) evaluate the strength of an *ideal* adversary. From an optimization aspect, Expression (9) indicates that the attack efficiency of an evidence-based distinguisher is related to the tightness of the identifiable leakage bound. The pathway for optimizing attack efficiency is clear: by approximating a tighter leakage bound, the efficiency of the distinguisher is coherently optimized. This philosophy resonates with the major contribution of this work. Since we have bridged the gap, it is advisable to exploit this established connection to optimize one aspect by improving the other.

5.2 How to Assess Worst-case Leakage Evidence

Based on Expression (2), assessing the worst-case leakage evidence is to determine a ζ_{wor} such that $\zeta_{wor} \leq \zeta$, which is equivalent to finding the infimum of ζ with respect to Θ, i.e.,

$$\inf_{\Theta} \zeta = \inf_{\Theta} \inf_{X} |\mathcal{D}(\mathcal{I}(F_*; \mathcal{X}_L); \Theta) - \mathcal{D}_{max/min}| \leq \zeta_{wor}. \tag{14}$$

This optimization rationale follows from the role of Θ. By analogy with the factor 2^t in the (t, δ)-SD-noisy leakage model of Crypto 24 [ORR+24], the embedding parameter Θ is a homogeneous factor used for model adaptivity. Notice that Expression (14) is not a closed form, we choose to use an optimization algorithm to make ζ asymptotically converge to ζ_{wor}. For the leakage evaluator $\mathcal{D}_{o_1}$ defined in Expression (10), the problem is modeled as

$$\begin{aligned} &\max_{\Theta} \mathcal{D}_{o_1}(\hat{\mathcal{I}}(F; \mathcal{X}_L); \Theta) \\ &\text{s.t.} \quad c > 0, \quad \sum_{|U_{i,j}|} \omega_{i,j} = 1, \quad \omega_{i,j} > 0, \quad l_j \in L/\{l_i\}. \end{aligned} \tag{15}$$

The difficulties in solving the problem (15) lie in: (I) the problem is subject to large-scale constraints, as each of the $|L|$ blocks is subject to one equality and one inequality constraint; (II) the problem is non-concave[5] with respect to the optimization matrix $diag(\omega_{i,j})$. To address these issues, we decompose the problem (15) into $(|L| - 1)$ subproblems and tackle them separately. Notice that the drift vector $\vec{C}$ is less sensitive to the matrix

[4]Here, "best achievable" refers to local optimality within the proposed attack, i.e., the best achievable performance relative to the attack itself, rather than global optimality among all possible attacks.

[5]It should be noted that although $\mathcal{D}_{o_1}$ is globally concave with respect to the nested $\widehat{\mathcal{I}}$, it may be non-concave when considered with respect to distinct variables.

$diag(\omega_{i,j})$, we set its entry c as a constant to reduce the overall constraint scale. Due to $\mathcal{D}_{o_1} = \sum_{l_j \in L/\{l_i\}} \mathcal{D}_{o_1}^{(i,j)}$, the relaxed subproblem associated with the l_i-th block is given by

$$\min_{\vec{w}} -\mathcal{D}_{o_1}^{(i,j)}, \quad \text{s.t.} \quad \sum_i \vec{w}\{i\} = 1, \quad \vec{w}\{i\} > 0, \quad i = 1, \cdots, |U_{i,j}|. \tag{16}$$

where the diagonal matrix $diag(\omega_{i,j})$ is equivalently replaced by the vector $\vec{w} = [\omega_{i,j}]^\top$ to reduce variable complexity, and the term $\vec{w}\{i\}$ denotes the i-th entry of the vector $\vec{w}$.

The Augmented Lagrangian Method (ALM) [Roc74] is an efficient algorithm for non-concave constrained problems, we thus adopt it as the basic framework for solving the relaxed subproblem given in Expression (16). Furthermore, considering the sensitivity of the objective function $\mathcal{D}_{o_1}^{(i,j)}$ to the optimization variable $\vec{w} = [\omega_{i,j}]^\top$, we integrate the Proximal Point Algorithm (PPA) [Gül91] into ALM to improve robustness. The augmented Lagrangian function associated with the subproblem in Expression (16) is given by

$$L_A(\vec{w}) = -\mathcal{D}_{o_1}^{(i,j)} + \mu_1(\sum_i \vec{w}\{i\} - 1) + \mu_2^\top \vec{w} + \frac{\beta}{2}(\sum_i \vec{w}\{i\} - 1)^2 + \frac{\gamma}{2}\|\vec{w}\|^2, \tag{17}$$

where μ_1 denotes a scalar-valued Lagrangian multiplier for the equality constraint, μ_2 denotes a vector-valued Lagrangian multiplier for the inequality constraint, and β, γ are scalar penalty parameters. Let $\zeta_{wor}^{(i,j)}$ denote the worst-case leakage evidence between the l_i and l_j blocks. We present the optimization algorithm in Algorithm 1 for approximating $\zeta_{wor}^{(i,j)}$. Let ω_t denote the iteration point at the t-th epoch. At the beginning of each epoch t, we first perform a steepest-descent step on the augmented Lagrangian function L_A in Expression (17) to derive an intermediate point $\vec{w}_{t'}$ at Step 4. Based on $\vec{w}_{t'}$, we apply the PPA to search for a proximal point $\vec{w}_t$ at Step 5. It is worth mentioning that $\vec{w}_t$ is an analytical solution to the minimization problem $\min_{\vec{w}}\{L_A(\vec{w}) + r\|\vec{w} - \vec{w}_{t'}\|^2\}$. By the first-order necessary condition of optimality [HY03], the iteration point $\vec{w}_t$ satisfies

$$\nabla L_A(\vec{w}_t) + 2r(\vec{w}_t - \vec{w}_{t'}) = \vec{0},$$

where $\vec{0}$ denotes an all-zero vector. Given that $\vec{w}_t$ is a proximation of $\vec{w}_{t'}$, by approximating the term $\nabla L_A(\vec{w}_t)$ by $\nabla L_A(\vec{w}_{t'})$, we derive the analytical solution $\vec{w}_t$ at Step 5 as

$$\vec{w}_t = \vec{w}_{t'} - \frac{1}{2r}\nabla L_A(\vec{w}_{t'}). \tag{18}$$

Finally, we conduct threshold projection and normalization at Steps 6, 7 to ensure that the conditions $\|\vec{w}_t\|_1 = 1$ and $\vec{w}_t > \vec{0}$ are consistently satisfied according to Proposition 1.

For other worst-case evidence $\zeta_{wor}^{(i,k)}$ with $l_k \in L/\{l_i, l_j\}$, we can identify each of them by applying Algorithm 1 to their corresponding relaxed subproblems. Note that for non-convex problems with linear constraints, the convergence rate of ALM is at least sublinear, therefore, the convergence rate at which $\zeta \to \zeta_{wor}$ is sublinear. In the security evaluation scenario where F_* is targeted as the key $\vec{K}_*$, the set $U_{i,j}$ used in Expressions (15), (16) and Algorithm 1 is replaced by the profiling set U_{pro} defined in Section 4.

The improved attack distinguisher $\mathcal{D}_{o_1}^{opt}(\hat{\mathcal{I}}(F; \mathcal{X}_L); \Theta_{opt})$ is derived by replacing the initialized factor Θ in $\mathcal{D}_{o_1}(\hat{\mathcal{I}}(F; \mathcal{X}_L); \Theta)$ by the optimized one $\Theta_{opt} = [diag(\vec{w}_T), \vec{C}]$ generated by Algorithm 1, where $diag(\vec{w}_T)$ represents the diagonal matrix whose main diagonal elements are filled by the entries of the vector $\vec{w}_T$ at the T-th iteration in Algorithm 1.

For high-order cases, approximating the worst-case leakage evidence is equivalent to solving optimization problems in which Expression (5) serves as the objective function. Since the corresponding high-order subproblems involve more complex vector and matrix operations, approximating the higher-order worst-case evidence is more challenging.

Algorithm 1: Optimization Algorithm for Solving the Subproblem (16)

Input: Collision set $U_{i,j}$, multipliers μ_1 and μ_2, penalty parameters β and γ, proximal parameter r, thresholding τ, step-size η, and iterations T.

Output: Worst-case leakage evidence $\zeta_{wor}^{(i,j)}$ between the l_i-th and l_j-th blocks.

Initialize $\vec{w}_0$ satisfying $\sum_i \omega_0\{i\} = 1$ and $\omega_0\{i\} > 0$, $t \leftarrow 0$;

while $t \leq T$ **do**

  Compute the gradient of L_A at the point $\vec{w}_t$, denoted by $\nabla L_A(t)$;

  Compute the gradient descent point: $\vec{w}_{t'} \leftarrow \vec{w}_t - \eta \nabla L_A(t)$;

  Compute the proximal point: $\vec{w}_t \leftarrow \mathrm{argmin}_{\vec{w}}\{L_A(\vec{w}) + r\|\vec{w} - \vec{w}_{t'}\|^2\}$;

  Conduct threshold projection for all $1 \leq i \leq |U_{i,j}|$: $\vec{w}_t\{i\} \leftarrow Proj(\vec{w}_t\{i\}, \tau)$,

$$Proj(\vec{w}_t\{i\}, \tau) = \begin{cases} \vec{w}_t\{i\} & \text{if } \vec{w}_t\{i\} \geq 0, \\ \tau & \text{if } \vec{w}_t\{i\} < 0, \end{cases}$$

  where $\vec{w}_t\{i\}$ denote the i-th entry of $\vec{w}_t$;

  Normalization: $\vec{w}_t \leftarrow \vec{w}_t / \|\vec{w}_t\|$;

  Record iterations: $t \leftarrow t + 1$;

Compute the value of $\mathcal{D}_{o_1}^{(i,j)}$ at the point $\vec{w}_T$, denoted by $\mathcal{D}_{o_1}^{(i,j)}(T)$;

return $\zeta_{wor}^{(i,j)} \leftarrow |\mathcal{D}_{o_1}^{(i,j)}(T)|$.

6 Experimental Analyses

6.1 Overview

In this section, we conduct four experiments to verify the practical applicability of the proposed leakage assessment model (Expression (2)), the characterized leakage evidence (Expressions (2) and (5)), and the evidence-based distinguishers (Expressions (10), (11), and (12)). These experiments will provide a thorough validation by covering the following aspects: (1) AES algorithm ($n = 8$) and PRESENT algorithm ($n = 4$); (2) simulated leakages and real-world leakages; (3) profiled attacks and non-profiled attacks; (4) two evaluation targets: F_* is either the collision chain or the key; (5) F_* is either fixed or random. The code is available at https://anonymous.4open.science/r/Evidence-Based-SCA-7E7D.

6.2 Experiments on Simulated Traces from AES-128

We generate simulated traces using the Hamming weight model $X_q^{(l_i)} = \mathrm{HW}(Sbox(t_q^{(l_i)} \oplus K_*^{(l_i)}) + \varepsilon$, where the leakage function is simulated as the composition of the Hamming weight function and the AES-128 S-box substitution. The noise term ε is modeled as a zero-mean random variable with variance σ^2. Therefore, the Signal-to-Noise Ratio (SNR) satisfies $SNR = np(1-p)/\sigma^2 = n/4\sigma^2$, where n denotes the input bit length of the S-box and $p = 0.5$ is the probability parameter in a binomial distribution of random plaintext.

To visualize the geometry of the proposed evaluator $\mathcal{D}_{o_1}$ given in Expression (10), we reduce its domain to a two-dimensional $dom(\mathcal{D}_{o_1}) = R^2$. To ensure the representativeness of this visualization, both the block pair (l_i, l_j) and the collision pair $(u_{i,j}, u'_{i,j})$ are randomly sampled from sets L and $U_{i,j}$, respectively. For the embedding factor $\Theta = [diag(\omega_{i,j}), \vec{C}]$, the constant entry c in the drift vector $\vec{C}$ is set to $c = 10^{-5}$, while the entries $\omega_{i,j}$ in the diagonal matrix $diag(\omega_{i,j})$ are drawn from a uniform distribution and normalized.

The visualization result is presented in Fig. 1. The full-leakage contour is delineated by a red dashed line (when adapted to the domain $dom(\mathcal{D}_{o_1}) = R^2$, the contour collapses to lines). The axes X and Y in the horizontal plane denote the two entries in the output

vector of the nested function $\hat{\mathcal{I}}$, that is, $(X, Y)^\top = \hat{\mathcal{I}}$. The vertical axis Z represents the value of $\mathcal{D}_{o_1}$, that is, $Z = \mathcal{D}_{o_1}$. The variance σ is simulated to $\sigma = 1, 2, 3, 4$, resulting in SNR ranging from 2 to 0.125. The number of traces $|Q|$ used in the assessment model is set to $|Q| = 120, 180, 200$. These combined settings lead to 12 simulated leakage scenarios, which are well differentiated by symbols in different colors and shapes.

The random geometric behavior of $\mathcal{D}_{o_1}$ confirms the topological coercivity condition and the global concavity property given in Theorem 2. Furthermore, the uniqueness of the contour confirms the conclusion of Lemma 1. As the SNR decreases or the number of traces $|Q|$ increases, the geometric points of $\mathcal{D}_{o_1}$ are projected into regions closer to the full-leakage contour, indicating that these points possess geometric properties more similar to those of the extremal points anchored on this contour. These observations confirm the interpretability mechanism established in Section 3.1. Therefore, the proposed interpretable assessment model satisfies the required conditions and is highly sensitive to varying leakage scenarios, we can thus confirm its effectiveness in practical applicability.

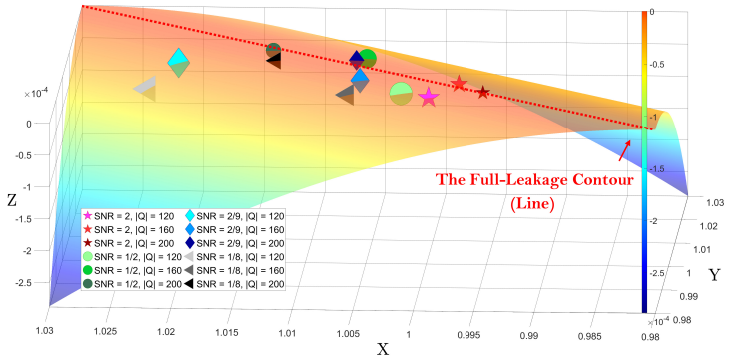


Figure 1: A side view of the random geometric behavior of the proposed evaluator $\mathcal{D}_{o_1}$ in Expression (10), adapted to the two-dimensional domain $dom(\mathcal{D}_{o_1}) = \mathbb{R}^2$. The markers on the random geometry denote different points of $\mathcal{D}_{o_1}$ under different $|Q|$ and SNR.

6.3 Experiments on the DPA Contest v4.1 Dataset

We use real traces from the DPA contest v4.1 dataset[6], which provides software implementations of the AES-256 algorithm protected by the rotating S-box masking (RSM). In this experiment, we target the security evaluation scenario where F_* corresponds to the collision chain and the proposed evidence-based distinguishers operate in non-profiled attacks. The embedding factor $\Theta = [diag(\omega_{i,j}), \vec{C}]$ is fine-tuned as follows. Each entry $\omega_{i,j}$ in the diagonal matrix $diag(\omega_{i,j})$ is assigned and normalized according to the average counts under the collision hypothesis, where these counts are estimated via uniform random sampling over the traces. The constant drift vector $\vec{C}$ is set with $c = 10^{-3}$.

The verification results are presented in Fig. 2. In the experiment, we vary the number of traces $|Q|$ and the trace batches to evaluate the multi-order leakage evidence, while estimating the empirical SR and GE metrics for the proposed evidence-based distinguishers. The value of $|Q|$ ranges from 60 to 2400 in increments of 60 traces, and for each fixed $|Q|$, the trace batches are randomly and independently sampled 1000 times. We use the natural logarithm to process these quantities for improved presentation of the curves.

We confirm that as the multi-order leakage evidence becomes more pronounced, i.e., as the evidence values decrease, the empirical SR increases while the GE correspondingly decreases. This monotonic trend across all Taylor orders is consistent with the analysis

[6] https://dpacontest.telecom-paris.fr/v4

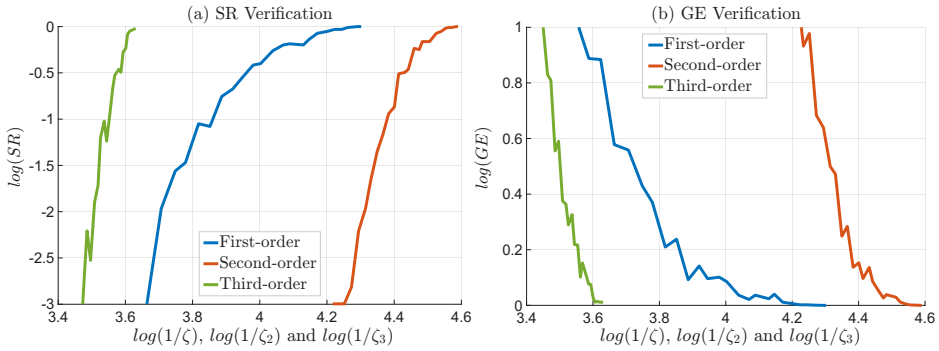


Figure 2: Verification results illustrating the correlation between multi-order leakage evidence ζ, ζ_2, ζ_3 and the attack metrics SR, GE on the DPA contest v4.1 dataset.

provided in Expression (9). This observation validates that the characterized multi-order leakage evidence and the common attack metrics admit practical mutual interpretation.

The optimization results are given in Fig. 3. In this experiment, we establish a concrete optimization scenario by fixing $|Q| = 1300$ and $(l_i, l_j) = (1, 16)$ based on Expression (16). In Fig. 3(a), three independent batches are sampled for repeated experiments. In Fig. 3(b), we apply cross-validation to improve the reliability of this verification. Specifically, we use 1000 different batches to compute empirical SR. To maintain consistent dimensions of the diagonal matrix $diag(\omega_{i,j})$, the collision set $U_{1,16}$ is sampled with identical cardinality and elements at each epoch. In the pre-processing stage, multiple preliminary runs of Algorithm 1 are executed to fine-tune the parameters as follows: $T = 400$, $\tau = 10^{-10}$, the combined step size $\eta_{com} = 10^{-5}$ (based on Expression (18) and Step 4 of Algorithm 1, the proximal parameter r and the step size η can be merged into a unified configuration), the combined penalty parameter $\beta_{com} = 10^{-5}$ (according to Expression (17)), $\mu_1 = 10^{-6}$ and $\mu_2 = [10^{-6}]^\top$. Furthermore, we implement an empirical initialization procedure to prevent the early optimization epochs from entering non-concave or non-convex domains.

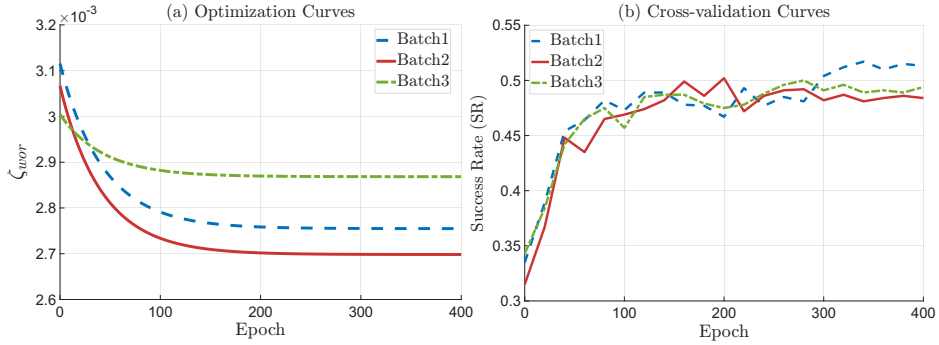


Figure 3: Optimization results illustrating the convergence toward the worst-case leakage evidence ζ_{wor} and the maximum SR on the DPA contest v4.1 dataset.

We confirm that the leakage evidence ζ in all three independent batches decreases monotonically and asymptotically converges to the worst-case leakage evidence ζ_{wor} as the optimization progresses. From the cross-validation curves in Fig. 3(b), we confirm that the empirical SR increases with slight oscillations, suggesting that the leakage evidence ζ connects the attack metric SR within a dynamic equilibrium. In particular, the proposed evidence-based distinguisher achieves the highest attack efficiency near the epochs at which the evidence ζ reaches its minimum (approximately around 200 epochs for Batch 1, 250 to 300 epochs for Batch 2, and 300 to 350 epochs for Batch 3). These observations

suggest that the leakage evidence can be used to assess the generalization of the proposed assessment model in terms of SR maximization and to determine an epoch threshold of early stopping for Algorithm 1. Furthermore, these observations corroborate the conclusion provided in Section 3.3 that the characterized evidence ζ quantifies exploitable leakages in the tailored attack $\mathcal{D}_{o_1}$, while non-exploitable counterparts are not identified by the evidence and therefore do not contribute to improving practical attack efficiency.

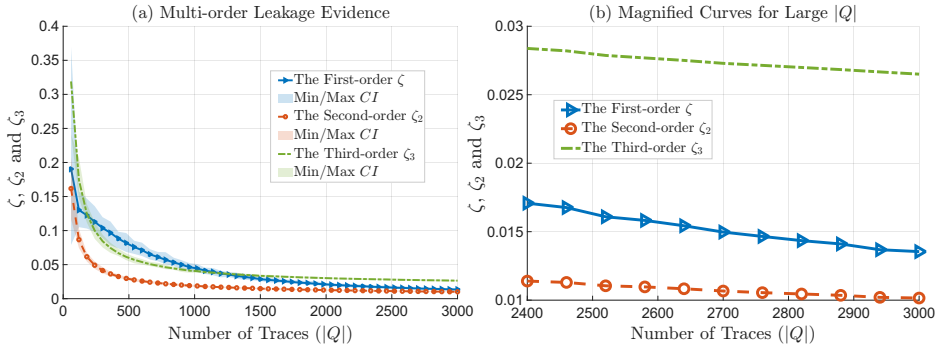


Figure 4: Leakage assessment results of the multi-order leakage evidence ζ, ζ_2, ζ_3 characterized by the proposed assessment model (1) on the DPA contest v4.1 dataset.

The leakage assessment results of the multi-order leakage evidence are given in Fig. 4. To quantify performance variability across repeated experiments, we depict the minimum and maximum confidence intervals (Min/Max CI) around the curves in Fig. 4(a). Fig. 4(b) magnifies the curves of Fig. 4(a) for large $|Q|$, specifically in the range 2400 to 3000.

The evidence values across all Taylor orders gradually fall into the same Landau order (up to constant factors), as observed in Fig. 4(a), thus validating Expressions (6) and (7) in Theorem 1. Furthermore, as $|Q|$ increases, the tangent slopes of the evidence curves become flatter and the Min/Max intervals narrower. This phenomenon is explained via information theory: when the sample size $|Q|$ is small, incorporating additional traces introduces non-redundant information, yielding wider intervals due to non-negligible bias across trace batches. In this regime, the leakage evidence is more sensitive to variations in the amount of information, leading to steeper slopes. For large $|Q|$, the added traces contribute mostly redundant information due to saturation. We ultimately confirm that these observations are consistent with the information-based findings observed in [DSP16] [CRBO24]. Thus, we can confirm the effectiveness of the proposed model.

6.4 Experiments on PRESENT on an ATmega328P Microcontroller

We use real traces from a software implementation of PRESENT running on an ATmega328P microcontroller operating at a 16 MHz clock frequency. The measurements were collected using a WaveRunner 8104 oscilloscope at a 1 GS/s sampling rate.

In the experiment, F_* is targeted as the collision chain and the evidence-based distinguishers operate in non-profiled attacks. The parameters follow the same configurations as in the previous experiments. The verification results are presented in Fig. 5.

We also confirm that as the evidence values decrease, the empirical SR increases while the GE decreases. This consistent trend again verifies that the characterized multi-order leakage evidence and the attack metrics SR and GE admit practical mutual interpretation. Unlike the validation curves shown in Fig. 2, the curves in Fig. 5 exhibit greater numerical stability. This can be attributed to the fact that the PRESENT algorithm uses an S-box with a smaller input bit length ($n = 4$) than that of the AES-256 algorithm ($n = 8$), resulting in real-world physical leakages with relatively lower noise variance.

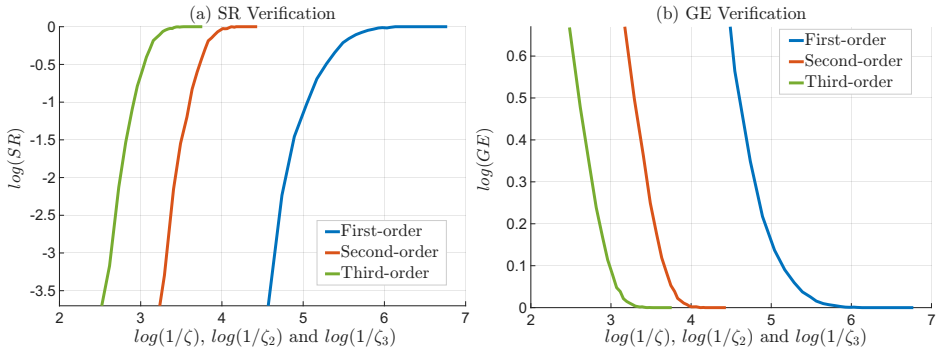


Figure 5: Verification results illustrating the correlation between multi-order leakage evidence ζ, ζ_2, ζ_3 and the attack metrics SR, GE on the PRESENT dataset.

The optimization results are presented in Fig. 6(a) and Fig. 6(b), respectively. In this experiment, we establish a concrete optimization scenario by fixing $|Q| = 48$ and $(l_i, l_j) = (1, 16)$ based on Expression (16). The parameters in Algorithm 1 are fine-tuned as follows: $T = 400$, $\tau = 10^{-10}$, the combined $\eta_{com} = 2 \cdot 10^{-4}$, the combined $\beta_{com} = 10^{-7}$, $\mu_1 = 10^{-6}$ and $\mu_2 = [10^{-6}]^\top$. We also implement an empirical initialization procedure to prevent the early optimization epochs from entering non-concave or non-convex domains. We confirm that as the leakage evidence ζ converges asymptotically toward the worst-case leakage evidence ζ_{wor}, the corresponding evidence-based distinguisher achieves an increasing SR in attack efficiency. This observation also validates that the characterized leakage evidence ζ connects the common attack metric SR within a dynamic equilibrium.

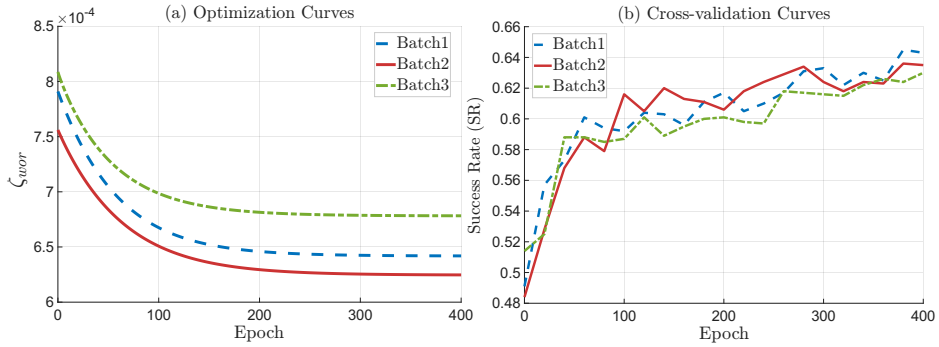


Figure 6: Optimization results illustrating the convergence toward the worst-case leakage evidence ζ_{wor} and the maximum SR on the PRESENT dataset.

6.5 Experiments on the (Random-Key) ASCADv1 Dataset

We use real traces from the (random-key) ASCADv1 dataset[7], a widely used benchmark for side-channel analysis on masked software implementations. For possible comparisons with parallel work, we follow the same convention in [BPS+20] by targeting the sensitive intermediate $Im = Sbox(t_q^{(l_3)} \oplus K_{rnd}) \oplus Mask_{out}$ and using traces from the third S-box. Here, K_{rnd} denotes the random key byte and the Boolean mask $Mask_{out}$ denotes an output mask employed to protect the recomputed S-box substitution. In this experiment, we target the security evaluation scenario where F_* corresponds to the key and implement profiled attacks. In the profiling stage, F_* is randomly provided, whereas during the attack

[7] https://github.com/ANSSI-FR/ASCAD

stage, F_* should be fixed. For each value of the sensitive intermediate Im, the leakage value is profiled as the simple average of 10 traces. The embedding factor $\Theta = [diag(\omega_{i,j}), \vec{C}]$ is fine-tuned as follows. Each entry $\omega_{i,j}$ in the sparse diagonal matrix $diag(\omega_{i,j})$ is assigned and normalized according to the average counts under the key hypothesis, where these counts are estimated via uniform random sampling over the traces. The constant drift vector $\vec{C}$ is set with $c = 10^{-3}$. The verification results are presented in Fig. 7.

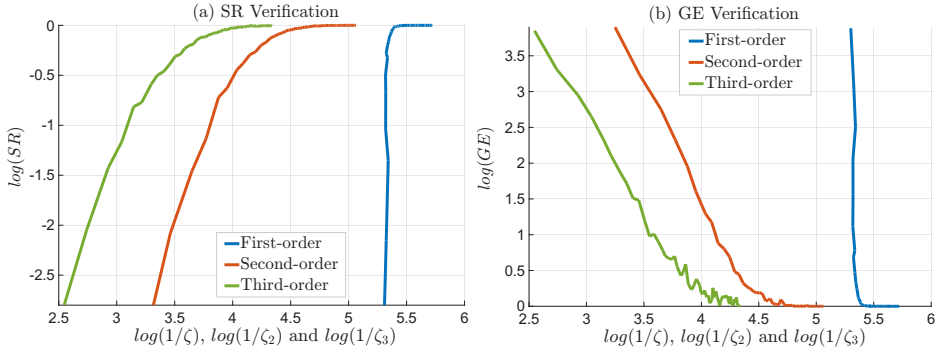


Figure 7: Verification results illustrating the correlation between multi-order leakage evidence ζ, ζ_2, ζ_3 and the attack metrics SR, GE on the (random-key) ASCADv1 dataset.

We observe a similar monotonic trend with slight oscillations in Fig. 7 on the (random-key) ASCADv1 dataset, where the empirical SR increases while the GE decreases as the multi-order leakage evidence becomes more pronounced. This consistent behavior across all three datasets confirms that the proposed assessment model connects the leakage evidence with the SR and GE attack metrics in an analytical manner under Expression (9).

In the optimization experiments, we establish a concrete scenario by fixing $|Q| = 20$ according to Expression (16). Multiple runs of Algorithm 1 are pre-executed to fine-tune the parameters as follows: $T = 600$, $\tau = 10^{-10}$, the combined $\eta_{com} = 10^{-4}$, the combined $\beta_{com} = 10^{-7}$, $\mu_1 = 10^{-7}$ and $\mu_2 = [10^{-7}]^\top$. We implement an initialization procedure to prevent the early optimization from entering non-concave or non-convex irregular domains. The optimization results illustrating the convergence toward the worst-case leakage evidence ζ_{wor} and the maximum SR are given in Fig. 8(a) and Fig. 8(b), respectively.

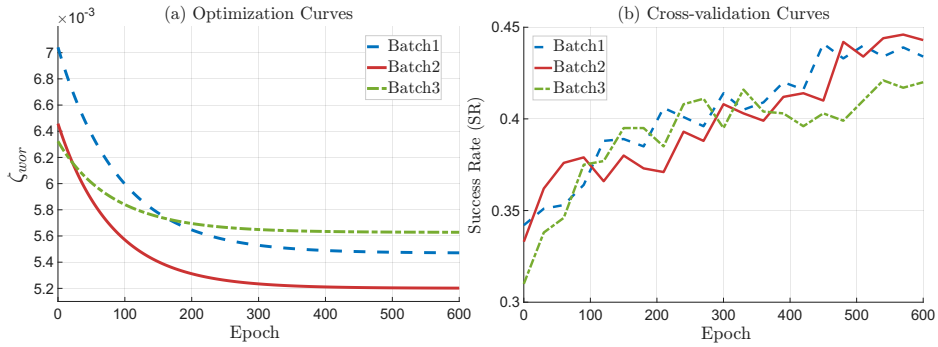


Figure 8: Optimization results illustrating the convergence toward the worst-case leakage evidence ζ_{wor} and the maximum SR on the (random-key) ASCADv1 dataset.

In Fig. 8(a), we observe consistent convergence behavior on the (random-key) ASCADv1 dataset. The leakage evidence ζ across all independent batches decreases monotonically and asymptotically converges to the worst-case leakage bounds ζ_{wor} as the optimization progresses. In Fig. 8(b), the cross-validation curves verify that tighter leakage bounds

approximated in later epochs uniformly lead to higher attack efficiency. The number of epochs required for final convergence differs (around 400 epochs for Batch 1, 300 epochs for Batch 2, and 350 epochs for Batch 3), but these differences are up to constant factors and thus share the same sublinear rate indicated in Section 5. In Fig. 8(a), it is worth noting that the leakage evidence in Batch 3, initially assessed as the strongest evidence, converges to the highest (i.e., weakest) worst-case bound among the three batches. This observation highlights the necessity of worst-case leakage assessment, as the worst-case leakage bound is not determined by the initial assessment. Furthermore, the optimization results indicate that Algorithm 1 jointly enables the proposed assessment model to approximate tighter leakage bounds for worst-case assessment, and the evidence-based distinguisher to achieve higher efficiency in attack evaluation. Thus, we can confirm the effectiveness of Algorithm 1.

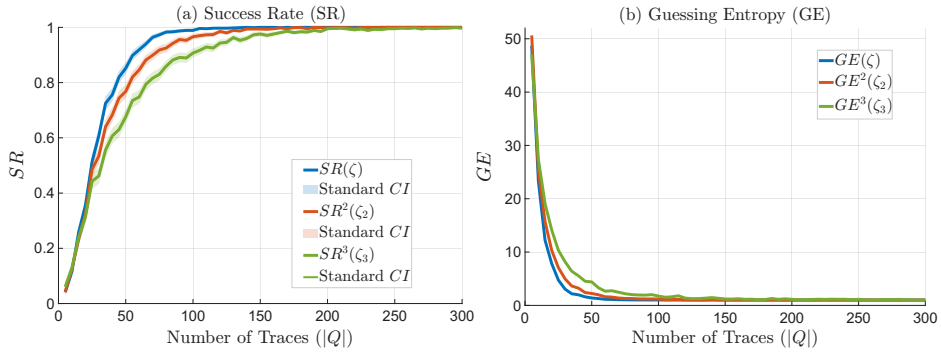


Figure 9: Attack evaluation results of the attack metrics SR and GE for the proposed evidence-based distinguishers $\mathcal{D}_{o_1}$, $\mathcal{D}_{o_2}$ and $\mathcal{D}_{o_3}$ on the (random-key) ASCADv1 dataset.

The attack evaluation results for the proposed evidence-based distinguishers are presented in Fig. 9. We denote the empirical SR and GE for $\mathcal{D}_{o_1}$ by $SR(\zeta)$ and $GE(\zeta)$, for $\mathcal{D}_{o_2}$ by $SR^2(\zeta_2)$ and $GE^2(\zeta_2)$, and for $\mathcal{D}_{o_3}$ by $SR^3(\zeta_3)$ and $GE^3(\zeta_3)$. The number of traces $|Q|$ ranges from 10 to 300. The standard confidence intervals (Standard CI) embedded around the SR curves denote the estimation error, namely, plus/minus one standard deviation $(SR(1-SR)/N_{\text{rep}})^{1/2}$, where $N_{\text{rep}} = 1000$ denotes the number of repeated experiments.

The empirical SR and GE for all evidence-based distinguishers converge to 1.00, as observed in Fig. 9(a) and Fig. 9(b), which validates the discrete sequence-limit analysis provided in Section 3.3. The successful attacks confirm that a concrete adversary can indeed exploit the GD-noisy leakage characterized by Definition 1 to recover F_*. It is also confirmed that the multi-order leakage evidence is derived correctly from F_*, rather than from irrelevant intermediates. In terms of attack efficiency, $\mathcal{D}_{o_1}$ outperforms $\mathcal{D}_{o_2}$, while $\mathcal{D}_{o_2}$ outperforms $\mathcal{D}_{o_3}$. This phenomenon can be explained by the dimensional expansion caused by Taylor's theorem, through which the leakage evidence evolves from scalar to vector and matrix representations. Higher-order leakage evidence thus contains richer elements. Since each element is uniformly affected by noise, the observed attack behavior is well explained. Practically, these results reveal a trade-off in tuning the Taylor order of the leakage evidence and the distinguisher: characterizing higher-order leakage evidence for richer information elements, or using lower-order distinguishers for higher attack efficiency.

7 Discussion

We provide several critical discussions concerning the proposed model and distinguishers:

Comparison with TVLA [GJJ+11] for leakage assessment. TVLA addresses the qualitative question of whether statistically significant leakages exist. As a hypothesis test, TVLA is prone to Type I and Type II errors and requires statistical corrections for error-rate

control [WO19]. In comparison, our model characterizes multi-order leakage evidence to address the quantitative question of the extent to which leakages reveal information of sensitive intermediates. Although the proposed model is subject to probabilistic errors, increasing the Taylor order of the leakage evidence (currently up to the third order) can exponentially reduce assessment errors and steadily improve model accuracy. Furthermore, attacks can be instantiated by anchoring the leakage evidence at its asymptotic minimum. In terms of their roles, the proposed model and TVLA may be viewed as complementary assessment frameworks, as they address different problems in leakage assessment.

Scalability of the proposed distinguishers. The scalability can be analyzed across three levels: the dimensionality of leakage variables, the convexity/concavity properties, and the coercivity condition required for geometric interpretability. At the dimensional level, the leakage variables used in Proposition 1 are currently vector-valued, and they can be reduced to a scalar representation or extended to a matrix representation by adjusting the nested function $\mathcal{I}$. At the functional level, the convexity/concavity properties used in Definition 1 can be relaxed to quasi-convexity/quasi-concavity to broaden the family of admissible functions, or strengthened to strict convexity/concavity to improve geometric regularity. At the topological level, coercivity can be generalized to other compactness conditions under Weierstrass theorem (Section 2, [BNO03]), such as bounded sublevel sets.

Limitations. Whilst the proposed model introduces an interpretable geometric paradigm for leakage assessment, certain aspects of its formulation may affect its applicability in specific scenarios. Since the model does not rely on distribution fitting, it avoids fitting errors; however, it may overlook subtle distributional features such as multi-modality or heavy tails. For instance, when leakages arise from a mixture of regimes (e.g., glitches, clock-domain crossings, or countermeasure phase changes), the resulting leakage distribution may become multi-modal, in which case a single geometric summary may underestimate the exploitability of a subset of traces. Similarly, heavy tails (rare but large-amplitude events) may dominate attacks even when the average-case leakage evidence appears modest. As a practical remedy, simple plug-in robustness options can be incorporated without sacrificing interpretability, such as outlier tracking. These diagnostics can help flag cases in which tails or mixture structures materially influence the results, and can further motivate robust variants of the proposed framework. From a Bayesian perspective, the model currently lacks a dedicated module for exploiting prior information. This module will be investigated in future work. Specifically, we plan to exploit prior information to develop a lightweight "Bayesian wrapper" around the evidence (which may technically yield posterior-weighted evidence) for leakage assessment and a MAP-style distinguisher for attack evaluation.

Comparison with commonly used attacks. Unlike conventional classifier-based (e.g., DPA [KJJ99] and PPA [LCC+06]) and predictor-based (e.g., LRA [MOW17]) attacks, the proposed distinguishers introduce a geometry-based attack mechanism to the SCA community. While this mechanism highlights the uniqueness of our attacks, their attack efficiency may not be optimal in specific scenarios. For instance, template attacks (TA) [CRR02] are theoretically information-optimal under Gaussian assumptions, and when optimal templates can be (approximately) profiled, their practical efficiency is likewise near-optimal [BCGR22]. It is worth noting that, in the context of interpretability studies, potential sub-optimality in attack efficiency is viewed as acceptable (e.g., see [GO24]). Compared with deep-learning based side-channel attacks (DL-SCAs) [WPP24] [Tim19] [MDP20] [WWK+23], the proposed distinguishers may require a higher level of expertise from the adversary. This is a common limitation inherent to analytical side-channel attacks in general, such as the information-theoretic MIA [GBTP08] and KSA [WOM11]. Still, it is important to acknowledge that state-of-the-art DL-SCAs may outperform our evidence-based distinguishers due to their robust feature extraction capabilities.

8 Conclusions

This work profiles an interpretable leakage assessment model based on a family of globally convex/concave and coercive functions. The mechanism of the model is grounded in geometry theory. We characterize exploitable multi-order leakage evidence, which enables us to develop three concrete evidence-based distinguishers. Furthermore, we propose an optimization algorithm to assess the worst-case leakage evidence. Importantly, this work helps bridge the long-standing gap between leakage assessments and attack evaluations, offering a new perspective for security evaluations. In future work, we will extend the proposed framework to higher-order (e.g., tensor-order) leakage evidence and their attacks.

Acknowledgements

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A Proof for Theorem 1

Proof. According to Taylor's theorem, we obtain the following expansion:

$$\begin{aligned}\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot) =& \mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta) + \nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)^\top(\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)) \\ &+ o(\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|).\end{aligned} \tag{19}$$

By straightforward rearrangement, Expression (19) can be rewritten as:

$$\begin{aligned}&|\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot) - \mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)| \\ &\leq |\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)^\top(\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L))| + o(\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|) \\ &\leq \|\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta) - \nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)\| \cdot \|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|,\end{aligned}$$

where we employ the triangle inequality and drop the high-order term $o(\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|)$. According to the first-order optimality condition $\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot) = \vec{0}$, we substitute the all-zero vector $\vec{0}$ with the gradient $\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)$. Also, the Cauchy-Schwarz inequality is employed in the second inequality. Thus, Expression (6) follows instantly from Expression (5). Moreover, applying Taylor's theorem to $\nabla\mathcal{D}$ yields:

$$\begin{aligned}\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot) =& \nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta) + \nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)^\top(\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)) \\ &+ o(\|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|).\end{aligned}$$

Similarly, applying the triangle inequality to the above expression yields:

$$\begin{aligned}&\|\nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot) - \nabla\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta)\| \\ &\leq \|\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{X}_L);\Theta) - \nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)\|_{\mathbf{F}} \cdot \|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\| \\ &+ \|\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)\|_{I_2} \cdot \|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\| \\ &= \mathcal{O}((\zeta_3 + \lambda_{max}[\nabla^2\mathcal{D}(\mathcal{I}(F_*;\mathcal{Z}_L);\cdot)]) \cdot \|\mathcal{I}(F_*;\mathcal{Z}_L) - \mathcal{I}(F_*;\mathcal{X}_L)\|) \simeq \mathcal{O}(\zeta_3),\end{aligned}$$

where the term $\|A\|_{I_2} = \max_{\|x\|\neq 0} \|Ax\|/\|x\| = \lambda_{max}[A]$ denotes the induced Euclidean norm with respect to the matrix A. This follows from the compatibility property between induced matrix norms and the corresponding vector norms (see, e.g., in [Dem97] [Axl24]). □